



Reflect with us - Week 4



This assignment will not be graded and is only for practice.

Find the dimension of the vector space

$$V = \{(x, y, z) \mid x, y, z \in \mathbb{R}, x + y + z = 0, z = 0\}$$

Addition: $(x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1 + x_2, y_1 + y_2, z_1 + z_2); \quad V = \{(x_1, y_1, z_1), (x_2, y_2, z_2) \in V$

Scalar multiplication: $c(x, y, z) = (cx, cy, cz); (x, y, z) \in V, c \in \mathbb{R}$

$\{(x, y, z) \mid x, y, z \in \mathbb{R}, x + y + z = 0, z = 0\}$ Addition: $(x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1 + x_2, y_1 + y_2, z_1 + z_2);$
 $(x_1, y_1, z_1), (x_2, y_2, z_2) \in V$ Scalar multiplication: $c(x, y, z) = (cx, cy, cz); (x, y, z) \in V, c \in \mathbb{R}$
 Solution: Solution:

Observe that, in question, you are asked to find the dimension of the vector space . So first you have to find a basis of the vector space.

Note: Note: Consider the system of linear equations

$$\begin{array}{rcl} x + y + z & = & 0 \\ z & = & 0 \end{array} \quad \begin{array}{rcl} x + y & = & 0 \\ x + y & = & 0 \end{array}$$

Find the solution of the system of linear equations.

Step 1: Step 1:

1 point

The system of linear equations has

- ☐ unique solution.
☐ no solution.
☐ infinitely many solution.

No, the answer is incorrect.

Score: 0

Feedback:

Use Gauss elimination method, and observe that $z = 0, z = 0$ and $x = -y, x = -y$.

Accepted Answers:

infinitely many solution.

1 point

Is $(1, -1, 0)$ is a solution of the system of linear equations?

- ☐ Yes
☐ No

No, the answer is incorrect.

Score: 0

Feedback:

Substitute $x = 1, y = -1, z = 0$ in both the equations of system of linear equations and check.

Accepted Answers:

Yes

1 point

Is $(1, -1, 0) \in V$?



☐ Yes

☐ No

No, the answer is incorrect.

Score: 0

Accepted Answers:

Yes

Claim: $\{(1, -1, 0)\}$ spans V .

Step 2:

1 point

Let $(a, b, c) \in V$. Which of the following options is/are true?

☐

$$a = -b$$

☐

$$a + b = 1$$

☐

$$c = 0$$

☐

$$c = 1$$

No, the answer is incorrect.

Score: 0

Feedback:

As vector (a, b, c) is a vector of the vector space then it will satisfy the system of linear equations $a + b + c = 0, c = 0$

Accepted Answers:

$$a = -b$$

$$c = 0$$

Hence, any vector $(a, b, c) \in V$ can be written as $(a, -a, 0)$.

1 point

Is $(a, -a, 0)$ a scalar multiple of $(1, -1, 0)$?

☐ Yes

☐ No

No, the answer is incorrect.

Score: 0

Accepted Answers:

Yes

1 point

If $(a, -a, 0) = \alpha(1, -1, 0)$, then what is the possible value for α ?

☐

$$a$$

☐

$$-a$$

☐

$$0$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$a$$

Hence, the set $\{(1, -1, 0)\}$ spans the vector space V .

Step 3: In this step we have to check whether the set $\{(1, -1, 0)\}$ is linearly independent or not.

1 point

Let S be a set which contains only one non-zero vector. Then which of the following options is true?

☐

S is linearly independent set.

☐

S is linearly dependent set.

No, the answer is incorrect.

Score: 0

Accepted Answers:

S is linearly independent set.

Recall: A basis of a vector space is a set of linearly independent vectors which spans the vector space.

From here you can conclude that $\{(1, -1, 0)\}$ is the basis of the vector space.

Step 4: In this step we have to find the dimension the vector space V .

Recall: The cardinality of the basis is the dimension of V .

Find the dimension of the vector space V .

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

[Check Answers](#)

Your score is: 0/8